

Identified hierarchical Bradley–Terry

for IPL match outcomes

A study guide, not a betting paper

Study desk notes for the `ipl-hierarchical` repo

September 2026

Abstract

This note is a study guide for the IPL hierarchical Bradley–Terry desk. The question is narrow: does an identified franchise-level logit, with only pre-match home and chase, give coherent *next-season* win probabilities that beat a coin and a pooled Bradley–Terry? The contribution is identification, a frozen predictor at forecast time, and a holdout that uses the franchise mean μ rather than a season effect that has not been observed. It is not a claim that the model beats a book.

A first 2026 holdout (72 decided matches, train $n = 1146$) returned Brier 0.263 against a coin’s 0.25, with mean predicted $p = 0.504$ against a team-1 base rate of 0.375. That is a result. This guide explains the model, the split, the scores, how to read that table, and what would have to change before the number can be promoted from “one split” to “the paper.”

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1 How to use this guide

Read it in this order if you are new to the repo:

1. Section 2 — what is allowed to count as a result.
2. Sections 4–10 — the model and the split.
3. Section 11 — Brier and log loss, not accuracy.
4. Section 13 — the number that is already on disk.
5. Section 20 — work the exercises on the real tape.

The code lives in `src/ipl.hier/`. The tape is `data/matches_raw.csv`. The fit script is `scripts/fit.py`. Compile this file with

```
cd docs && pdflatex research_note.tex && pdflatex research_note.tex
```

This is a study desk. It is not a production tip sheet. If a sentence would only make sense as betting advice, it does not belong here.

2 The question

Let $y_m = 1$ if the designated team 1 wins IPL match m . We want $P(y_m = 1)$ for matches in a season that was *not* used to fit season-specific strengths.

A model is admissible only if all three hold:

1. The same linear predictor η is used to rank teams and to form $P = \text{logit}^{-1}(\eta)$.
2. Franchise strengths are identified: $\sum_i \mu_i = 0$.
3. Holdout probabilities do not use α_{i,s^*} for an unseen season s^* .

Do not. A large posterior gap in μ printed as $p = 0.51$ is not a forecast. It is a train/predict mismatch. The NFL companion project did that. This repo exists so that mismatch cannot hide in an IPL table.

2.1 What would count as a result

- Holdout Brier materially below 0.25 *and* below a pooled identified Bradley–Terry fit only on train.
- Intervals for μ that separate a small set of franchises from zero, with expansion sides covering zero.
- β_c near zero would mean the chase flag in *this* specification is not doing work. Report that. The opposite (chase away from zero, home near zero) is also a result.
- Failure relative to a closing line, if one is added later, is a result. Do not put the line inside η .

2.2 What would not count

Playoff 0–1 counts. In-sample accuracy. A large α gap printed as $p = 0.51$. Any table that uses α_{i,s^*} on the holdout. A 70/30 split of the official holdout used to fit Platt or a superlearner.

2.3 The paper is not “add features until Brier drops”

IPL squads turn over every auction. A player embedding, an XI list, or a pile of last-five run rates can cut in-sample Brier and still be illegal at forecast time if you do not have those columns on next year’s unplayed tape. The design choice is: only things you know before the first ball of an unseen season, plus a franchise label.

3 Why not a Gaussian run margin

A companion NFL desk uses a Gaussian point margin because that league’s score difference is closer to that story. Copying it here is the same mistake in the other direction.

T20 is two innings of twenty overs. The second innings is a chase with a known target. DLS rewrites the target when rain enters. Super overs break ties. None of that is a single Gaussian $y = \text{runs}_1 - \text{runs}_2$. The honest observation for “who won the match” is Bernoulli.

Binary winner throws away margin information. That is a cost. The cost of the alternative is a likelihood that is not the data. This desk pays the first cost on purpose.

4 Bradley–Terry from scratch

Definition 4.1 (Bradley–Terry). Two players i and j have positive strengths π_i, π_j . Then

$$P(i \text{ beats } j) = \frac{\pi_i}{\pi_i + \pi_j}.$$

Take $\pi_i = e^{\theta_i}$. Then

$$P(i \text{ beats } j) = \text{logit}^{-1}(\theta_i - \theta_j).$$

That is the whole model, before hierarchy and covariates. Team 1 beating team 2 in match m is

$$y_m \sim \text{Bernoulli}(\text{logit}^{-1}(\theta_1 - \theta_2)).$$

Fact 4.1 (Monotonicity). P is strictly increasing in the strength gap $\theta_1 - \theta_2$. $P = 1/2$ if and only if $\theta_1 = \theta_2$. A gap of 2 on the logit scale is already $P \approx 0.88$.

That identity is the point of the repo. If the table ranks i far above j and then prints $P \approx 1/2$, the printed P did not come from the same η that produced the rank.

4.1 Why team 1 / team 2 is a label, not a fact

Cricsheet lists two sides as `team1` and `team2`. That order is not “home” and not “bats first.” The observation $y = 1$ means “the listed team 1 won.” Strengths are still identified from differences. What is *not* identified without extra structure is an intercept that says “whoever is listed first wins more.” If a later season lists the eventual winner second more often, mean predicted p can sit at 0.50 while $\bar{y}$ sits at 0.38. That is Section 13.

5 Identification

Add a constant c to every θ_i . Differences $\theta_i - \theta_j$ do not change. The likelihood is unchanged. The posterior on θ without a constraint is a ridge, not a hill. Samplers wander. Tables become unreadable (“CSK is +40, MI is +39” instead of “CSK is +0.4 above average”).

Rule 5.1 (Sum-to-zero). This desk identifies franchise means by

$$\mu_i = \mu_i^{\text{raw}} - \bar{\mu}^{\text{raw}}, \quad \sum_i \mu_i = 0.$$

In code that is a **Deterministic** on top of an unconstrained **Normal**. Do not drop the centering.

A corner constraint ($\mu_{\text{MI}} = 0$) also identifies the model. Sum-to-zero is nicer to read: zero is “league average franchise,” not “Mumbai.”

Season effects $\alpha_{i,s}$ are used only as differences $\alpha_{1,s} - \alpha_{2,s}$ inside a season. A common intercept added to every side in season s cancels. You do not need a second sum-to-zero inside each season for the likelihood. You might still want one for interpretation of $\alpha_{i,s}$ as “this franchise in this year, relative to the field.”

6 The hierarchical model

For match m in season s , listed team 1 vs listed team 2:

$$\eta_m = \alpha_{1,s} - \alpha_{2,s} + \beta_h h_m + \beta_c c_m, \quad (1)$$

$$y_m \sim \text{Bernoulli}(\text{logit}^{-1}(\eta_m)). \quad (2)$$

Layer the season strengths on a franchise mean:

$$\alpha_{i,s} \sim N(\mu_i, \sigma_\alpha^2), \quad (3)$$

$$\mu_i \sim N(0, 1) \quad \text{with} \quad \bar{\mu} = 0, \quad (4)$$

$$\beta_h, \beta_c \sim N(0, 0.5^2), \quad (5)$$

$$\sigma_\alpha \sim \text{HalfNormal}(0.5). \quad (6)$$

In code, (3) is non-centered: $\alpha_{i,s} = \mu_i + \sigma_\alpha \cdot z_{i,s}$, $z \sim N(0, 1)$. That is the parameterization NUTS prefers when σ_α is small.

6.1 What each piece is for

Symbol	Role	Plain language
μ_i	franchise mean	Long-run label: “MI,” not “MI’s 2024 XI.”
$\alpha_{i,s}$	season strength	That label plus a year shock.
σ_α	year-to-year scale	How much a franchise wobbles around μ .
h_m	home flag	+1 if team 1 is at a mapped home city, −1 if team 2 is, else 0.
c_m	chase flag	1 if team 1 bowls first / chases.
β_h, β_c	logit shifts	How many logits home or chase is worth.

6.2 Holdout predictor

For an unseen season s^* the desk replaces $\alpha_{\cdot, s^*}$ by μ :

$$\eta_m^{\text{OOS}} = \mu_1 - \mu_2 + \beta_h h_m + \beta_c c_m. \quad (7)$$

Home and chase are legal: you know the city and the toss before the first ball. A new α_{i, s^*} is not legal. Drawing it from $N(\mu_i, \sigma_\alpha^2)$ and then *conditioning on 2026 results* would be fitting the holdout. Drawing it from the prior and averaging is almost μ anyway when σ_α is small.

6.3 What is not in η

Betting odds. EPA-style box-score piles. Player lists. Injury lists. Pitch reports. Clamped coefficients. A week AR(1). A stadium random effect is optional (`--venue`) and is not the headline model.

Do not. Closing prices may be scored as a *rival forecast* (`src/ipl_hier/market.py`). They must never become a column in η . If the book is inside the predictor, you are no longer asking whether franchise μ forecasts the next season. You are asking whether you can shrink toward a market that already saw the toss and the weather.

7 Priors, in units you can feel

On the logit scale, 0.5 is already a large match effect. If $\beta_c = 0.5$ and the game is otherwise even,

$$P = \text{logit}^{-1}(0.5) \approx 0.62.$$

The $N(0, 0.5^2)$ prior on β says “home and chase are probably smaller than that, and could be zero.” It is informative on purpose. IPL seasons are short. An uninformative prior on two extra logits will wander.

$\mu_i \sim N(0, 1)$ after centering is a wide franchise prior. A one-logit franchise is $P \approx 0.73$ against an average side with no home or chase. Historical IPL dynasties are not larger than that for long. $\sigma_\alpha \sim \text{HalfNormal}(0.5)$ says year shocks are probably smaller than a full logit. The 2026 fit put $\hat{\sigma}_\alpha \approx 0.13$, which is consistent with that prior being loose rather than binding.

8 Covariates

8.1 Home

$h_m \in \{-1, 0, 1\}$ comes from a city→franchise map in `src/ipl_hier/names.py`. Cities that do not map (UAE bubble, South Africa 2009, most dual-home experiments) stay 0.

The prose in the original short note said playoffs are also 0. The code, as first shipped, does *not* zero playoffs in a mapped city. A final in Ahmedabad still gives Gujarat Titans $h = +1$ if they are team 1. That is a doc/code mismatch. If you tighten the rule, put it in `clean_matches` and add a test. Do not leave it as a comment.

Home is a city lookup, not a stadium random effect and not a travel model. Ahmedabad→GT for every year is harmless when GT is not playing (the flag never matches either side). It is not a history of who hosted that ground.

8.2 Chase

$c_m = 1$ if team 1 is the side batting second. The tape has toss winner and toss decision; the chasing side is determined from those two fields. Neutral venues still have a chasing side. Super overs are dropped with ties.

Historically, listed team 1 chases much less than half the time on this tape (around 0.24 after cleaning). The chase coefficient is still identified. It is just identified off a minority of rows. Do not read β_c as “the league-wide P(chase wins).” Read it as “the logit shift when the listed team 1 is the chasing side.”

9 The tape

`data/matches_raw.csv` is parsed from Cricsheet IPL male CSV2 info files, 2007/08 through 2026. Ball-by-ball files are not stored. Rebuild with

```
python scripts/build_matches.py --zip ipl_male_csv2.zip
```

9.1 Cleaning rules

- Ties and no-results are dropped. Decided matches only.
- $y = 1$ iff `winner == team1`.
- Name continuity: Delhi Daredevils $\rightarrow$ Delhi Capitals, Kings XI $\rightarrow$ Punjab Kings, RCB Bangalore $\rightarrow$ Bengaluru.
- Defunct sides stay distinct: Deccan, Kochi, Pune Warriors, Gujarat Lions, Rising Pune.

Merging Daredevils into Capitals is a continuity choice, not a fact about the 2012 roster. Deccan $\neq$ SRH is the opposite choice on the same kind of city. Both are defensible. Changing them changes μ for those labels. Write the choice down before you refit.

9.2 What the snapshot looks like

On the snapshot that produced the 2026 holdout:

Raw rows	1243
Decided rows after clean	1218
Train rows (seasons before 2026)	1146
Holdout rows (2026)	72
Seasons on the tape	19

`event` is Indian Premier League on every row, so it cannot mark playoffs. `match_number` missing is a usable playoff-ish proxy (finals and qualifiers), not a formal BCCI flag.

10 Train / test protocol

Rule 10.1 (Last-season holdout). Train on every decided match with season key strictly before the holdout year. Test the holdout year. Teams that do not appear in train are dropped from test (expansion debut has no μ).

`holdout_last_season` sorts seasons by the integer before `/`, so 2007/08 precedes 2009, and a named `--holdout-season 2025` also drops 2026 from train.

One last-season split is the minimum publishable design. It is not the last word. Rolling origin — train through t , test $t + 1$, for several t — is the next experiment. A single 72-row season cannot separate “ μ does not transfer” from “2026 was a weird labeling year.”

10.1 Why not random row splits

Matches in a season share a squad, a pitch slate, and an auction. A random 80/20 split leaks 2026 information into μ through other 2026 rows. Time-respecting splits are the only splits this desk treats as holdout.

10.2 Why not use last year’s α as next year’s α

You can *blend* P_μ with P built from the last observed season’s α , with a weight λ chosen on earlier seasons. That is a second-stage layer (`src/ipl_hier/stack.py`). It is not the headline model. Choosing λ on the official holdout is cheating.

11 Scores

Let $p_m \in (0, 1)$ be the posterior mean of $\text{logit}^{-1}(\eta_m)$ under the frozen predictor.

Definition 11.1 (Brier).

$$\text{Brier} = \frac{1}{n} \sum_m (p_m - y_m)^2.$$

A coin ($p = 1/2$) scores 0.25 on any binary tape. Lower is better.

Definition 11.2 (Log loss).

$$\text{LogLoss} = -\frac{1}{n} \sum_m (y_m \log p_m + (1 - y_m) \log(1 - p_m)).$$

A coin scores $\log 2 \approx 0.693$. Lower is better. A single overconfident miss is expensive.

Accuracy $\frac{1}{n} \sum_m \mathbf{1}\{p_m > 0.5\} = y_m$ is reported and is not used to rank models. On a tape with $\bar{y} = 0.375$ and $\bar{p} = 0.50$, accuracy near 0.47 is not a mystery and not a leaderboard.

11.1 How to think about Brier

The useful comparisons for this desk:

- A coin always scores 0.25.
- A constant forecast equal to the *holdout* base rate r scores $r(1 - r)$. For $r = 0.375$ that is 0.234. You cannot use the holdout rate as a model — you would be fitting the test set — but it tells you how much room a calibrated constant would have had.
- A constant forecast equal to 0.50 when $r = 0.375$ already costs about 0.017 extra Brier from calibration alone.

If holdout Brier is 0.263 with $\bar{p} = 0.50$ and $r = 0.375$, a large piece is “the model did not know team 1 would win 38% of the time.” The rest is ranking error.

11.2 Why Brier, not “we went 10–4 in playoffs”

Knockout matches are a handful of Bernoulli draws. They are fine as anecdotes and illegal as a ranking criterion. Brier uses every decided row and penalizes both arrogance and cowardice.

12 How to read a report

After

```
PYTHONPATH=src python scripts/fit.py --data data/matches_raw.csv --out reports
```

you have three files.

12.1 reports/metrics.json

Quote holdout Brier and log loss. Compare them to 0.25 and 0.693. Compare mean holdout p to the holdout base rate. If those two diverge by more than a few points, the model is miscalibrated on the team 1 label before it has a chance to be a good ranker.

`in_sample_*` will look better. That is the in-sample glow. Ignore it in the abstract.

As first shipped, `fit.py` does *not* write coin or pooled BT into this file even though `baselines.py` exists. Until that is wired, compute those two numbers by hand before you talk about “the hierarchy helping.”

12.2 reports/franchise_strength.csv

Rows are μ_i posterior means and 5–95% intervals, sorted. Intervals that cover 0 are “not separated from average.” Expansion sides and one-season guests should cover 0. If they do not, the prior on μ is too tight or the label merge is wrong.

Check $\sum_i \hat{\mu}_i \approx 0$ on the means. It will not be machine zero in a CSV of rounded floats, but it should be noise.

12.3 reports/holdout_YEAR.csv

One row per holdout match: `p_team1`, 5–95% bands, the raw columns. Bands are posterior uncertainty on p , not “this game is a 90% lock because the interval is narrow.” A narrow interval around 0.51 is a confident coin.

12.4 Coefficients

- $\beta_h \approx 0$ with a posterior that includes zero: the city map is not doing work in this spec. Could be a true small home edge in IPL. Could be a bad map.
- β_c away from zero: chase, as encoded, is doing work. Report the sign and the implied probability shift at a 50/50 game, $\text{logit}^{-1}(\beta_c) - 1/2$.
- σ_α small: season shocks are small, so OOS μ is close to what you would have used in-sample. If OOS still fails, the problem is not “we dropped α_{s*} .” The problem is that μ itself did not travel.

13 The 2026 holdout that already exists

One fit on this tape produced:

	In sample	Holdout 2026
n	1146	72
Accuracy	0.595	0.472
Brier	0.237	0.263
Log loss	0.666	0.719
Base rate $\bar{y}$	—	0.375
Mean p	—	0.504
$\hat{\beta}_h$		0.096
$\hat{\beta}_c$		0.353
$\hat{\sigma}_\alpha$		0.128

13.1 How to say this in one paragraph

An identified hierarchical Bradley–Terry with pre-match home and chase, predicting 2026 from franchise μ only, had holdout Brier 0.263 against a coin’s 0.25 and log loss 0.719 against $\log 2 \approx 0.693$. Mean predicted p was 0.50 while listed team 1 won 38% of decided matches. Chase was not idle ($\hat{\beta}_c \approx 0.35$, about +8 to +9 points at a 50/50 game). Home was small ($\hat{\beta}_h \approx 0.10$). Season wobble was small ($\hat{\sigma}_\alpha \approx 0.13$). In-sample Brier 0.237 is not the result.

13.2 What that does and does not mean

It does *not* mean IPL match outcomes are a coin. It means this predictor, on this split, did not beat $p = 1/2$.

It does *not* mean the hierarchy is worthless. We still lack the pooled-BT number on the same 72 rows. If pooled BT is also ≈ 0.26 , the extra season layer is not the villain. The 2026 team 1 base-rate shift is.

It does *not* license putting $\alpha_{i,2026}$ back into (7). That would fit the holdout.

Seventy-two binaries are a small test. A crude standard error on Brier is on the order of 0.03. 0.263 versus 0.25 is not a precise “worse than a coin.” It is a precise “did not materially beat a coin.” That is enough to refuse a victory lap and not enough to abandon rolling origin.

13.3 A short calibration identity

A constant forecast $p = 0.504$ against $r = 0.375$ has Brier

$$(1 - r)p^2 + r(1 - p)^2 \approx 0.251.$$

The observed 0.263 is that miscalibration plus about 0.01 of extra error. Most of the loss versus a coin is the team 1 rate, not a handful of playoff upsets.

14 Baselines

A referee will ask for these. None may use $\alpha_{i,2026}$.

Name	Construction
Coin	$p = 1/2$. Brier 0.25, log loss $\log 2$.
Always team 1	$p = 1$. Accuracy equals the holdout base rate.
Train rate	Constant $p = \bar{y}_{\text{train}}$. Legal.
Pooled BT	Identified logit, no season layer, coefficients frozen from train.
Ablation	Same hierarchy with $\beta_h = \beta_c = 0$.

Pooled BT must be fit to convergence, not one IRLS step from $p = 1/2$. `baselines.pooled_bt_eta` as first shipped is the one-step sketch. Do not quote it as MLE.

Rank models by holdout Brier and log loss. Report accuracy in a footnote.

15 Second-stage layers

`stack.py` has Platt scaling, a residual logit on home/chase, a λ -blend of P_μ and last-season P_α , and a discrete nonnegative superlearner. All of these are legal only if the weights are fit on seasons *before* the season you score.

Do not. `scripts/stack_holdout.py` as first shipped fits Platt on the first 70% of the official holdout file and scores the last 30%. That is not a 2026 confirmation. Either pass an earlier season’s holdout file as the calibration set, or do not run the script.

16 Computation and diagnostics

The sampler is PyMC NUTS, not a hand-rolled Gibbs sampler. Defaults in `fit.py` are two chains, 800 tune, 800 draws, `target_accept=0.9`, seed 42.

What belongs in a supplement, if you save `idata`:

- $\hat{R}$ on $\mu, \beta_h, \beta_c, \sigma_\alpha$. Fail the run if $\max \hat{R} > 1.02$.
- Bulk ESS on those same names. A few hundred is acceptable for a desk; a few dozen is not.
- Divergences. Non-zero at 0.9 accept usually means the non-centered α is still unhappy. Raise accept, or switch the unused team-season cells to a ragged index.

As first shipped, `fit.py` discards `idata` after writing CSV summaries. You cannot audit a fit you did not save.

The dense α array is shaped (teams $\times$ seasons), including Kochi in 2024 and CSK in a suspended year. Those cells have prior and no likelihood. They waste sampler time and smear σ_α only weakly. A ragged index of observed (i, s) pairs is the clean repair.

17 Limits (pre-registered)

- Franchise label is not a roster. Auction turnover is absorbed into σ_α and into holdout error.
- Home is a city lookup. Dual homes, playoffs, and relocated games are either 0 or silently wrong, depending on the map.

- Two never-overlapping franchises share information only through other μ 's. Expansion sides are noisy.
- Binary winner ignores margin. A 1-run choke and a 60-run thrashing update α the same way.
- Team 1 is a Cricsheet listing. A season that lists winners second will look like a base-rate shock.
- One last-season holdout is not rolling origin.

18 Relation to the NFL repair

Same discipline, different likelihood. The NFL branch is a Gaussian margin with optional week AR(1). This branch is a season-layered logit. Both refuse collinear box-score piles. Both freeze the predictor at forecast time. A joint paper is possible. A joint design matrix is not.

The failure mode this desk is designed to forbid is specific: a fitted margin model whose forecast path drops the covariates that made the margin large, then prints a 50/50 win probability next to a huge mean margin. Equation (7) is the IPL version of “the forecast path is the training path, minus information you do not have yet.”

19 Suggested next experiments

In order. Do not skip to a neural net.

1. Wire coin, train-rate, and converged pooled BT into `fit.py`. Write them into `metrics.json`.
2. Zero h_m on playoff-ish rows, or change the prose. Add a test.
3. Save `idata` and $\hat{R}$.
4. Score 2026 after flipping each row so p is $P(\text{favorite wins})$. If that Brier falls under 0.25, the wound is the team 1 label.
5. Rolling origin, 2019 through 2026, same predictor.
6. Estimate a λ -blend of P_μ and last observed P_α on seasons before the test year.

Closing-line comparison is allowed only as a rival column, and only with n reported.

20 Exercises

Work these on the real tape. They are the study guide.

Exercise 20.1 (Identity). Take $\eta \in \{-3, -1, 0, 1, 3\}$. Compute $\text{logit}^{-1}(\eta)$ by hand. Confirm $P(0) = 1/2$ and $P(2) > 0.88$. This is `tests/test_identities.py`.

Exercise 20.2 (Sum to zero). Draw a vector $\mu^{\text{raw}} \sim N(0, 1)^{15}$. Form $\mu = \mu^{\text{raw}} - \bar{\mu}^{\text{raw}}$. Check $\sum \mu = 0$. That is the identifier in `model.build_model`.

Exercise 20.3 (Split hygiene). Call `holdout_last_season` with season 2025. Assert that train contains no 2025 and no 2026 rows, and that every train season key is < 2025 .

Exercise 20.4 (Coin). On the 2026 holdout file, compute Brier of $p = 1/2$ (must be 0.25) and Brier of $p = \bar{y}_{2026}$ (must be $r(1 - r)$). The second number is illegal as a model and legal as a ceiling.

Exercise 20.5 (Favorite flip). On each 2026 row let $p' = \max(p, 1 - p)$ and $y' = y$ if $p \geq 1/2$ else $1 - y$. Compute $\text{Brier}(y', p')$. Compare to 0.263.

Exercise 20.6 (Chase shift). Using $\hat{\beta}_c = 0.353$, compute $\text{logit}^{-1}(0.353) - 1/2$. That is the implied shift at an even game when listed team 1 chases, holding μ and home fixed.

Exercise 20.7 (Why in-sample looks good). Brier in sample was 0.237 with season-specific $\alpha_{i,s}$. Brier out of sample was 0.263 with μ only. Write two sentences on why that gap is expected and why it is not evidence that $\alpha_{i,2026}$ should be used on 2026.

21 Glossary

η Linear predictor. Ranks and probabilities must share it.

μ_i
Identified franchise mean. Holdout strength.

$\alpha_{i,s}$
Season strength. Illegal on an unseen season.

σ_α
Scale of year shocks around μ .

h_m
Home edge in $\{-1, 0, 1\}$.

c_m
Chase flag for listed team 1.

Brier

Mean squared error of p on $y \in \{0, 1\}$. Coin = 0.25.

Log loss

Mean negative log likelihood. Coin = $\log 2$.

Holdout

Last season, or season $t + 1$ after training through t .

Rival forecast

A p you score and never put in η .

22 Code map

Path	Role
<code>src/ipl_hier/names.py</code>	Aliases and city→home map.
<code>src/ipl_hier/data.py</code>	Clean, encode, split.
<code>src/ipl_hier/model.py</code>	PyMC model and in-sample p .
<code>src/ipl_hier/predict.py</code>	OOS p from μ ; rank table.
<code>src/ipl_hier/metrics.py</code>	Brier, log loss.
<code>src/ipl_hier/baselines.py</code>	Coin, team 1, pooled BT.
<code>src/ipl_hier/market.py</code>	Decimal odds → rival p .
<code>src/ipl_hier/stack.py</code>	Platt / blend / superlearner.
<code>scripts/fit.py</code>	Headline train/test.
<code>scripts/report_team.py</code>	Filter, not a one-team model.
<code>scripts/stack_holdout.py</code>	Second stage; check the split.
<code>scripts/build_matches.py</code>	Cricsheet zip → CSV.
<code>tests/</code>	Identities and cleaning smoke tests.

References

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